

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Invoice Processing Agent | Banking & Finance Domain

Detail	Description
Audience	Engineering & Arts students (industry-aligned challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — mirrors reference career-advisor pattern

1. Problem Statement

Banks and finance departments receive a high volume of vendor invoices by email every day, each varying in format, currency, and payment terms. Accounts Payable teams currently open every email, read the invoice, verify vendor and amount details against purchase orders, check for duplicate or overdue invoices, and decide whether to approve, flag for query, or route to a specialist. This manual triage is time-consuming, delays vendor payments, and is prone to human error during month-end closing periods when invoice volume peaks.

Design and build an Autonomous UiPath Agent, the AI Invoice Processing Agent, that reads an incoming vendor invoice email, reasons over its content, classifies it, extracts structured billing details, and independently completes the required action end-to-end — without any human approval step — by invoking RPA workflows and Generative AI activities as tools.

2. System Prompt

You are an AI Invoice Processing Agent for the Accounts Payable department.
Your goal is to analyze vendor invoice emails and decide the best possible action.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - New Invoice Submission
 - Payment Reminder
 - Duplicate Invoice
 - Other
3. If it is New Invoice Submission, Payment Reminder, or Duplicate Invoice:
 - Extract vendor name, invoice number, amount, and due date
 - Decide if it is urgent (e.g., due date within 3 days or overdue)
 - If urgent, recommend immediate logging and notification
 - Generate a professional acknowledgement reply
4. If it is Other:
 - Skip processing

```
Rules:
- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details
- Return output in structured format
```

3. User Prompt

```
Analyze the following vendor invoice email and decide the appropriate action:
{{invoice_email_text}}
```

4. Input Schema

```
{
  "invoice_email_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "vendor_name": "string",
  "invoice_number": "string",
  "amount": "string",
  "due_date": "string",
  "urgency": "string",
  "action": "string"
}
```

6. Tool: Build Workflow & Publish

Create the following process inputs for the RPA workflow the agent will invoke:

- category (text)
- vendor_name (text)
- invoice_number (text)
- amount (text)
- due_date (text)
- urgency (text)

7. Add Tool: Map Inputs

Agent Field	Process Argument
category	in_category
vendor_name	in_vendor_name
invoice_number	in_invoice_number
amount	in_amount

Agent Field	Process Argument
due_date	in_due_date
urgency	in_urgency

8. Updated System Prompt — Tool Invocation Logic

If the email is classified as New Invoice Submission, Payment Reminder, or Duplicate Invoice:

- Use the log_to_excel tool to store the extracted invoice details in the Accounts Payable register.
- Use the generate_reply GenAI activity to draft a professional acknowledgement email to the vendor.
- If urgency = "High", use the notify_finance_team RPA workflow to alert the concerned finance officer automatically (fully autonomous, no manual approval required).

If the email is classified as Other:

- Take no further action.

9. Tools & Activities Summary

Tool / Activity	Type
log_to_excel	RPA Workflow (UiPath Excel Automation)
generate_reply	GenAI Activity (LLM prompt-based text generation)
notify_finance_team	RPA Workflow (Outlook / Teams notification)
classify_invoice	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent completes classification, extraction, logging, notification, and reply generation without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a multi-step reasoning system prompt for an autonomous agent
- Define structured input/output schemas for agent-to-workflow communication
- Map agent-extracted fields to RPA process arguments
- Combine RPA workflows and GenAI activities as callable tools within one agent
- Publish and test a complete autonomous agent in UiPath Agent Builder